

Q1

A person has three coins with two possible outcomes for each, a 'head' or a 'tail'. The person will toss each coin one after the other, that is, Coin1, Coin2, Coin3. We assume that the person will enter a value of 1 for a 'head' and 0 (zero) value for a 'tail'.

Your task is to **write pseudocode and draw flowchart** for an application which displays "You Won!" if the person gets two heads on any two consecutive coins and display "You Lose" otherwise. You cannot use logical AND/OR for any purpose.

Example1: (bold values are inputs)

Enter Coin1 value: **1**

Enter Coin2 value: **1**

Enter Coin3 value: **0**

You Won!

Example2: (bold values are inputs)

Enter Coin1 value: **0**

Enter Coin2 value: **1**

Enter Coin3 value: **0**

You Lose

- 1 - Start
- 2 - twoHeads = 0
- 3 - Print "Enter Coin1 value: "
- 4 - Input Coin1
- 5 - Print "Enter Coin2 value: "
- 6 - Input Coin2
- 7 - ~~Print~~ "Enter Coin3 value: "
- 8 - Input Coin3
- 9 - If Coin1 == 1
 - 9.1 - If Coin1 == Coin2
 - 9.1.1 - twoHeads = 1
- 10 - else if Coin2 == 1
 - 10.1 - If Coin2 == Coin3
 - 10.1.1 - twoHeads = 1
- 11 - If twoHeads == 1
 - 11.1 - Print "You Won!"
- 12 - else
 - 12.1 - Print "You Lose"
- 13 - Halt

(Start + 3 inputs)

02 points

(Coin1 & Coin2 heads)

02 points

(Coin2 & Coin3 heads)

02 points

(Msg. for ~~any~~ only one heads pair)

02 points

(Msg. on lose)

015 points

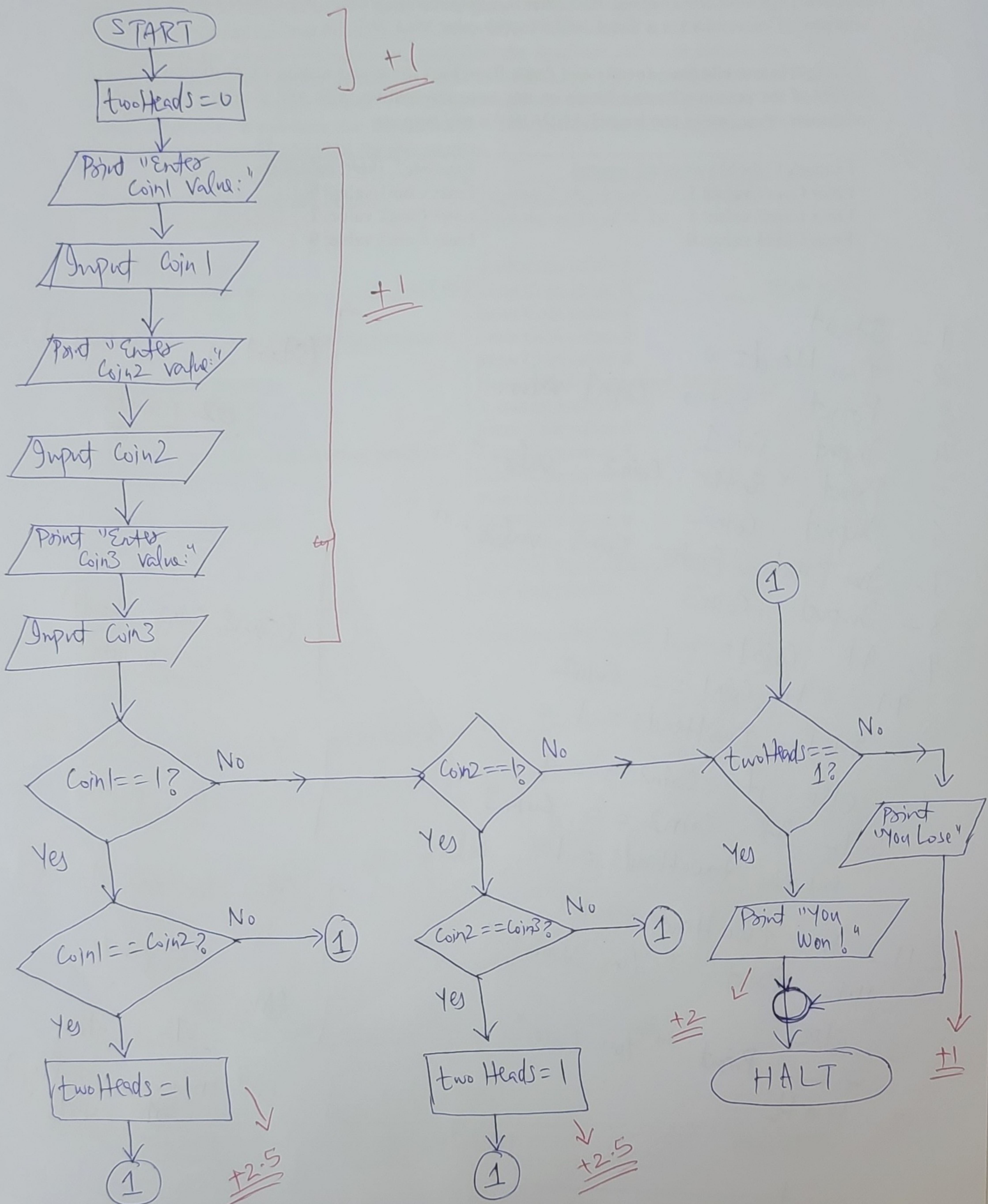
(All ending logically)

015 points

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A person will toss one coin N number of times and will enter a value of 1 for a 'head' and 0 (zero) value for a 'tail'. Your task is to **write pseudocode** for an application which finds a combination of *head-tail-tail* followed by its reverse, *tail-tail-head*. It is not necessary that both combinations are placed consecutively, however, the *head-tail-tail* must come first. Also, display the starting and ending position of each combination (see the examples below). Display a message, "Double combo found!" if both combinations are found and display "No combo found" otherwise. We assume that the person will always enter the value of N greater than 5. You cannot use logical AND/OR for any purpose.

<u>Example1: (bold values are inputs)</u> Enter the value of N: 6 Enter Coin value: 1 Enter Coin value: 0 Enter Coin value: 0 Head-Tail-Tail appeared: starting position = 1, ending position = 3 Enter Coin value: 0 Enter Coin value: 0 Enter Coin value: 1 Tail-Tail-Head appeared: starting position = 4, ending position = 6 Double combo found!	<u>Example2: (bold values are inputs)</u> Enter the value of N: 11 Enter Coin value: 1 Enter Coin value: 0 Enter Coin value: 1 Enter Coin value: 0 Enter Coin value: 0 Head-Tail-Tail appeared: starting position = 3, ending position = 5 Enter Coin value: 1 Enter Coin value: 1 Enter Coin value: 0 Enter Coin value: 0 Enter Coin value: 0 Enter Coin value: 1 Tail-Tail-Head appeared: starting position = 9, ending position = 11 Double combo found!
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- 1- Start
 - 2- prevCoin1 = -1, prevCoin2 = -1, Coin = -1, HTT = 0, TTH = 0,
start HTT = -1, end HTT = -1, start TTH = -1, end TTH = -1.
 - 3- Print "Enter the value of N:"
 - 4- Input N
 - 5- count = 1
- +1 (necessary variables, N input, count variable)

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6- while Count $\leq$ N +1 (loop & counter update)

6.1- Print "Enter Coin Value:"

6.2- Input Coin +1 (only one input inside the loop)

6.3- If HTT == 0 +2 (check if HTT not found)

6.3.1- If Coin == 0

6.3.1.1- If prevCoin1 == 0

6.3.1.1.1- If prevCoin2 = 1

6.3.1.1.1.1- If prevCoin2 = 1

6.3.1.1.1.1.1- HTT = 1

6.3.1.1.1.1.2- prevCoin2 = 1

6.3.1.1.1.3- end HTT = count

6.3.2- If HTT != 1

6.3.2.1- prevCoin2 = prevCoin1

6.3.2.2- prevCoin1 = Coin

6.4- else if TTH == 0 +1

6.4.1- If Coin == 1

6.4.1.1- If prevCoin1 == 0

6.4.1.1.1- If prevCoin2 == 0

6.4.1.1.1.1- If TTH = 1

6.4.1.1.1.1.1- prevCoin2 = 1

6.4.1.1.1.1.2- end TTH = count

6.4.1.1.1.3- end TTH = count

6.4.2- If TTH != 1

6.4.2.1- prevCoin2 = prevCoin1

6.4.2.2- prevCoin1 = Coin

6.5- count = count + 1

7- If HTT == 0

7.1- Print "No Combo found"

8- else if TTH == 0

8.1- Print "No Combo found"

9- else

9.1- Print "Double Combo found"

10- Halt

(unnecessary) ←

(check HTT combination, nested + three variables) +2

(Reset variables) +2

start HTT = count - 2 +1

(Update variables on condition check) +2

(check TTH) +1

(update variables on condition check) +2

(for each point) +1

03